

Programming Using C++ Level 1

Relational Operators



Programming Using C++

Level 1

Relational Operators

Definition

- ◆ Relational operators are used in C++ to compare two values.
- ◆ The result of a relational operation is a Boolean value:
 - 1 means True.
 - 0 means False.

Explanation

◆ This program asks the user to enter two numbers, *A* and *B*, and then compares them using different relational operators.

Steps in main()

1 Declare and read the first number

- ◆ The program creates an integer variable called *A* and asks the user to enter its value.

2 Declare and read the second number

- ◆ The program creates an integer variable called *B* and asks the user to enter its value.

3 Equal to ==

- ◆ Checks whether *A* and *B* have the same value.

→ Result:

- ◆ True if they are equal.

- ◆ False if they are different.

4 Not equal !=

- ◆ Checks whether *A* and *B* have different values.

→ Result:

- ◆ True if they are different.

- ◆ False if they are equal.

5 Greater than >

- ◆ Checks whether *A* is greater than *B*.

→ Result:

- ◆ True if *A* is greater than *B*.

- ◆ False otherwise.

Steps in main()

6. Less than <

◆ Checks whether A is less than B.

➔ Result:

◆ True if A is less than B.

◆ False otherwise.

7. Greater than or equal >=

◆ Checks whether A is greater than or equal to B.

➔ Result:

◆ True if A is greater than B or if both values are equal.

◆ False otherwise.

8. Less than or equal <=

◆ Checks whether A is less than or equal to B.

➔ Result:

◆ True if A is less than B or if both values are equal.

◆ False otherwise.

💡 Example

◆ If $A = 10$ and $B = 5$:

▣ $A == B \rightarrow 0$

▣ $A != B \rightarrow 1$

▣ $A > B \rightarrow 1$

▣ $A < B \rightarrow 0$

▣ $A >= B \rightarrow 1$

▣ $A <= B \rightarrow 0$

➡ Result

◆ The program prints 1 for True and 0 for False.

Note

- ◆ The program prints relational results as 1 or 0 by default.
- ◆ 1 means True.
- ◆ 0 means False.
- ◆ `#include <cmath>` is not needed in this program because no functions from the `cmath` library are used.

⊘ Warning

- ◆ Do not confuse = with ==.
- ◆ = assigns a value, while == compares two values.



Comparison



=

- ✓ Assignment operator.
- ✓ Assigns a value to a variable.
- ✓ Example: `A = 10`



==

- ✓ Relational operator.
- ✓ Compares two values.
- ✓ Example: `A == B`

✦✦ Summary

- ✓ Relational operators compare two values.
- ✓ The main relational operators are `==`, `!=`, `>`, `<`, `>=`, and `<=`.
- ✓ The result of a relational comparison is True or False, represented by 1 or 0.

Thank you 😊